EARTH'S PAST

The landscape features we see on Earth today are the result of events and processes stretching back more than 4 billion years. It is only in the last few hundred years, however, that scientists have pieced together some of the main details of the story.

△ CRYSTAL EVIDENCE
A zircon crystal, from Jack Hills, Australia, has been dated at 4.375 billion years old and is the oldest known material of terrestrial origin.

Earth's age and origins

A precursor body of what is now planet Earth formed around 4.55 billion years ago, from collisions of smaller bodies in a disk of material spinning around the Sun. When it was perhaps 40 million years old, the proto-Earth is thought to have experienced one final collision, leading to the formation of what we now call Earth and its natural satellite, the Moon.

The early Earth was probably born in an extremely hot, molten state. Heavier materials, mainly iron, sank to the center, with lighter materials forming layers around this. For around the first 150 million years, no solid crust formed because there were continuing impacts from comets and asteroids, while high volcanic activity continually reworked the surface. Around 4.37 billion years ago, the oceans had begun to form, through condensation of water released into the atmosphere by ancient volcanoes. By 4 billion years ago, the first pieces of continental crust had also formed. Movements of early tectonic plates caused landmasses to collide and merge, gradually forming the ancient cores of today's continents.

◁ ANCIENT COLLISION
A collision between a proto-Earth and a Mars-sized body, some 4.51 billion years ago, is thought to have led to the formation of the Earth-Moon system.

The geological timescale

The fact that Earth is extremely ancient, and that the sedimentary rock layers (strata) of its continental crust were laid down in sequence over a vast period of time, first became clear to scientists between the 17th and 19th centuries. They noticed that many rock strata contain fossils—the remains of ancient, apparently extinct, animals and plants. They divided the fossil-bearing rock strata—and thus Earth's history as a life-bearing planet—into three eras: the Paleozoic (meaning ancient life), Mesozoic (middle life), and Cenozoic (recent life). Later, these eras were subdivided into geological periods. In rock strata deeper and older than the fossil-bearing rocks, there seemed at first to be no life. These rock strata were later subdivided into three extremely long time